Convergence and Divergence in Predictive Maintenance

An engineering systems perspective



PRESENTER _____ • NOV 5, 2026

Learning objectives & roadmap

01 Understand maintenance fundamentals and predictive workflows

02 Explore technology convergence across the stack

03 Examine divergence by asset, failure mode and strategy

04 Apply concepts to a motor/pump and ESP32 prototype

1 Maintenance Fundamentals

2 Convergence Concepts

3 Divergence Concepts

4 Use Case & Architecture

5 Prototype & Roadmap

Maintenance keeps assets useful–

Maintenance

Activities performed to preserve or restore an asset’s required functionality.

Operational reliability

Production efficiency

Safety & lifecycle value



Three maintenance logics, three risk profiles

REACTIVE

Fix after failure

\+ Low planning effort

— High downtime & risk

PREVENTIVE

Service on schedule

\+ Predictable work

— May replace healthy parts

PREDICTIVE

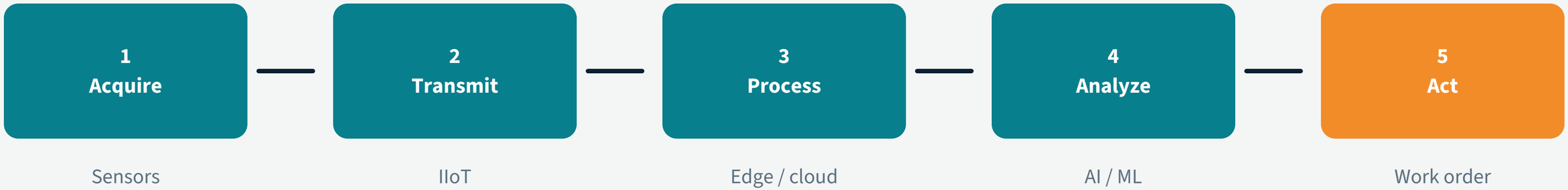
Act on condition

\+ Fewer surprises

— Needs data & skills

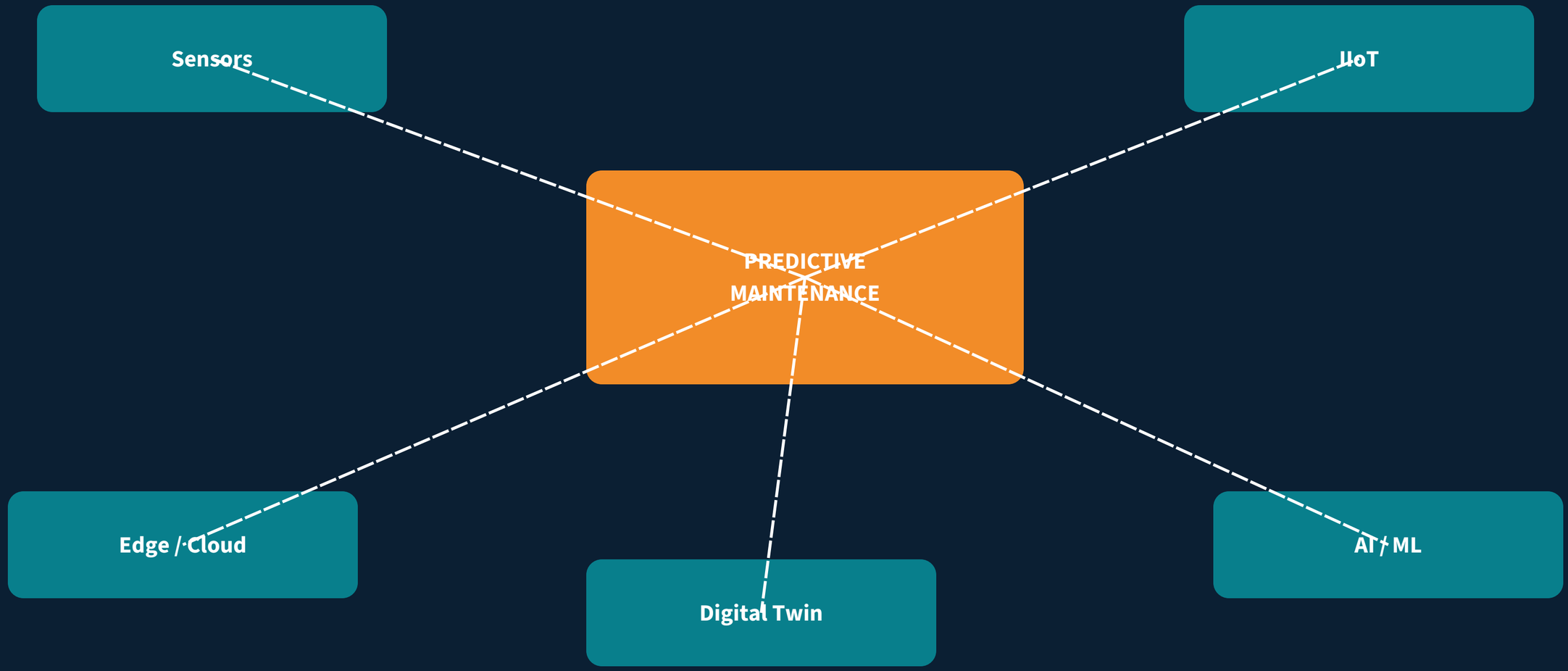
Predictive maintenance targets the intervention window: not too early, not too late.

Predictive maintenance is a closed decision loop

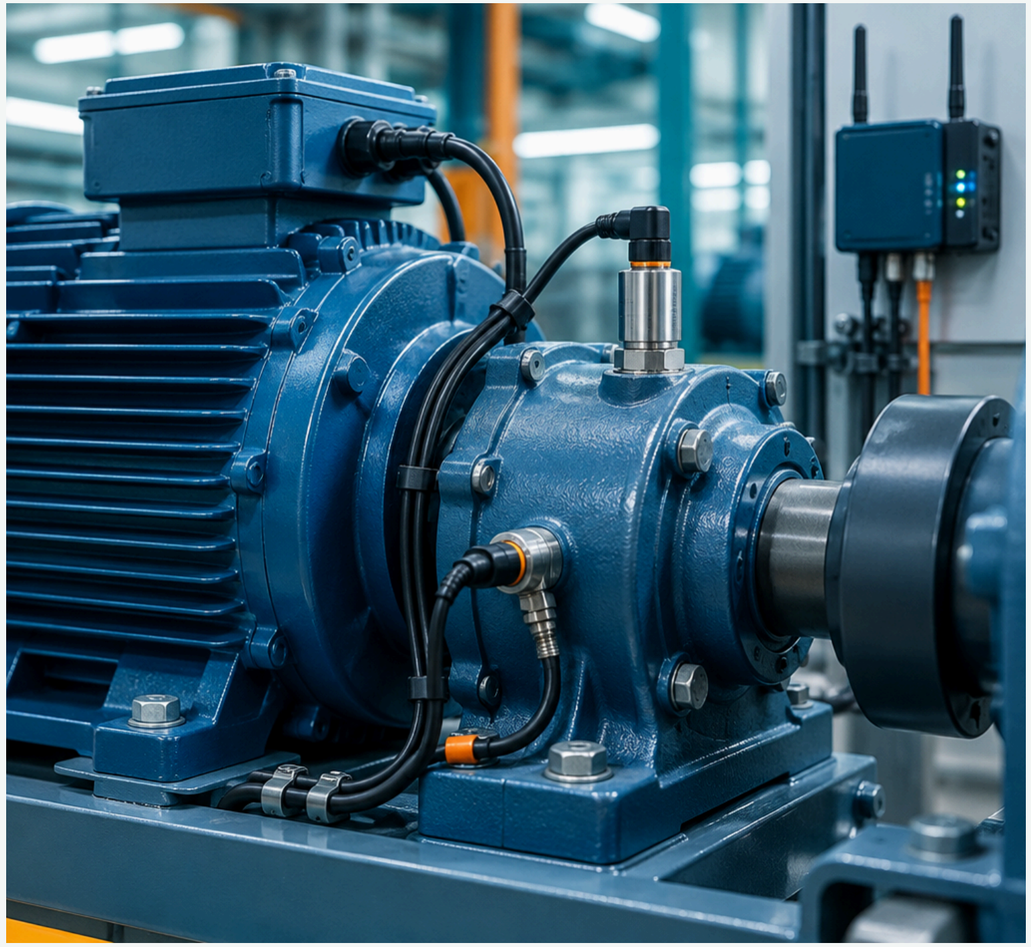


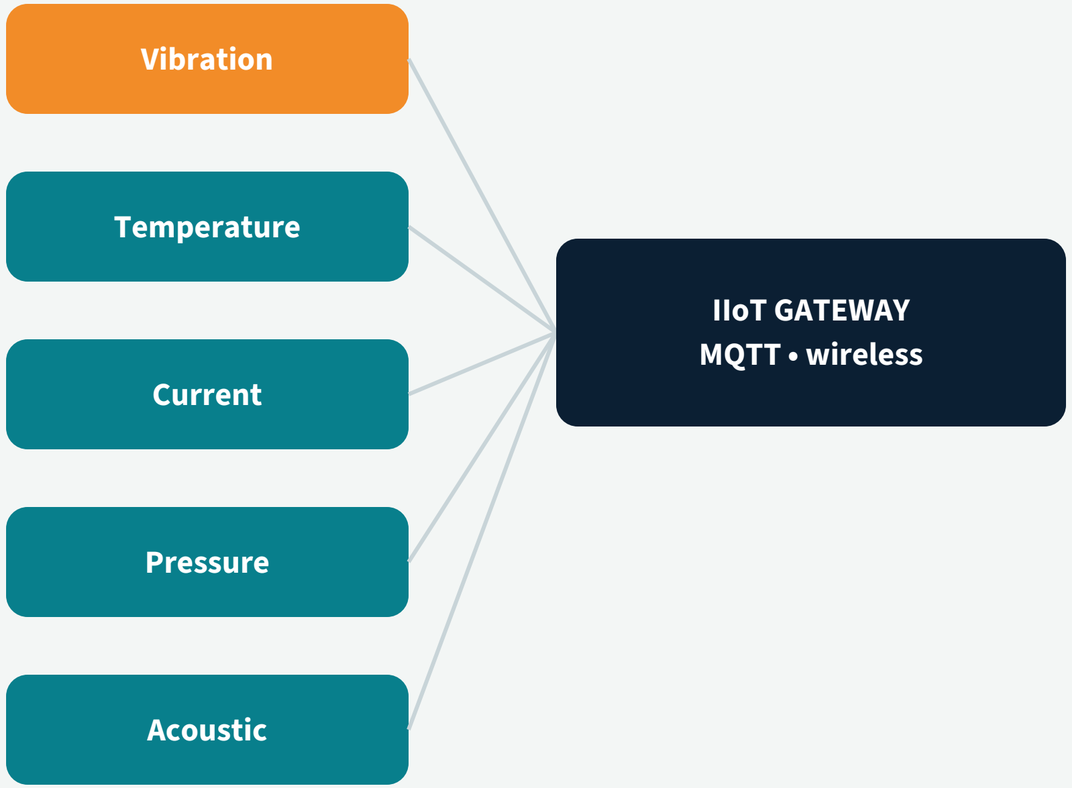
Condition data → evidence → timely intervention

Convergence turns components into a maintenance system



Sensors create the industrial nervous system





Place computation where it creates the most value



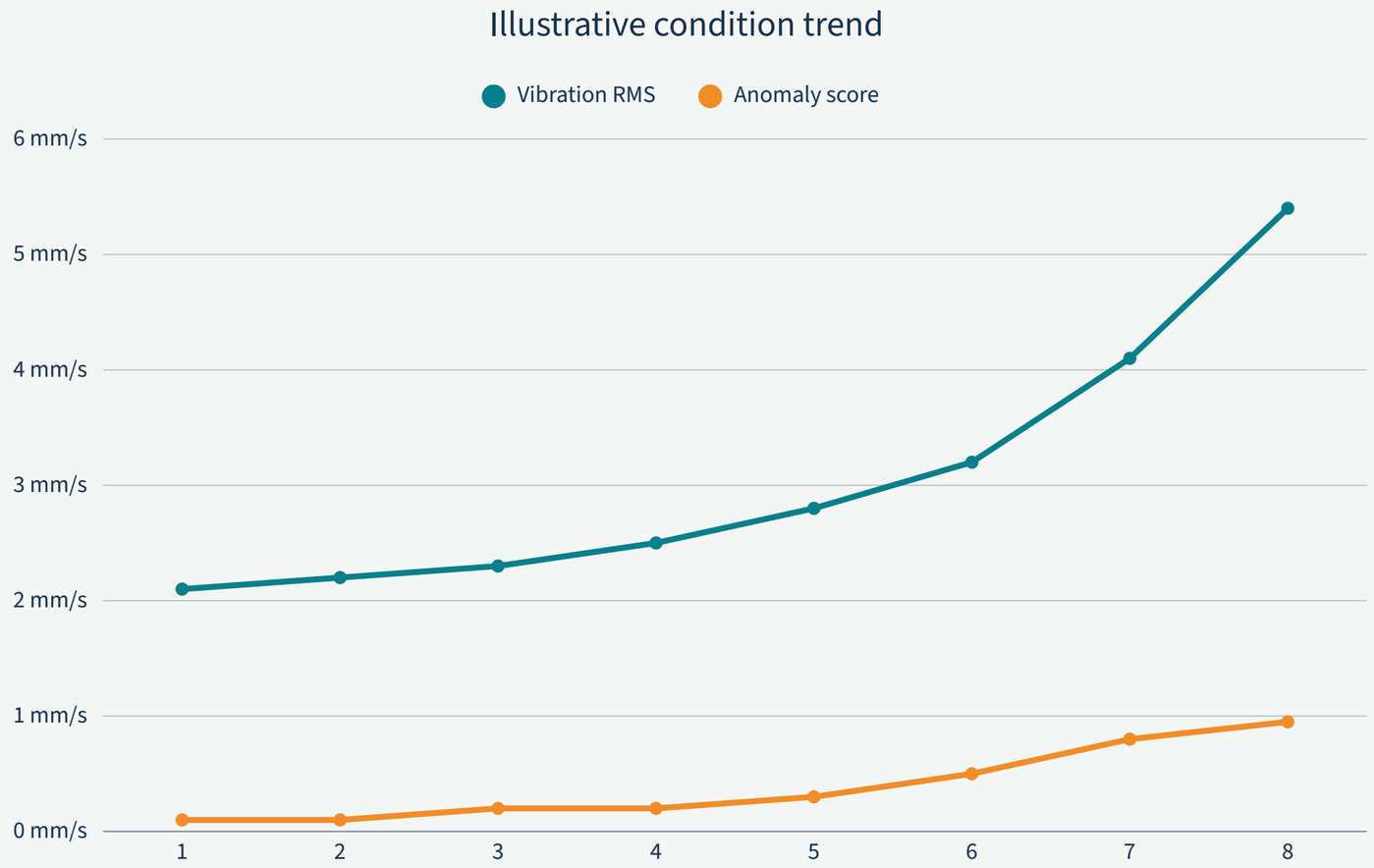
EDGE PRIORITY

Latency, bandwidth, resilience and local safety context.

CLOUD PRIORITY

Long horizons, cross-asset learning and scalable compute.

AI/ML converts condition signals into foresight



DETECT

Anomaly detection

DIAGNOSE

Failure classification

FORECAST

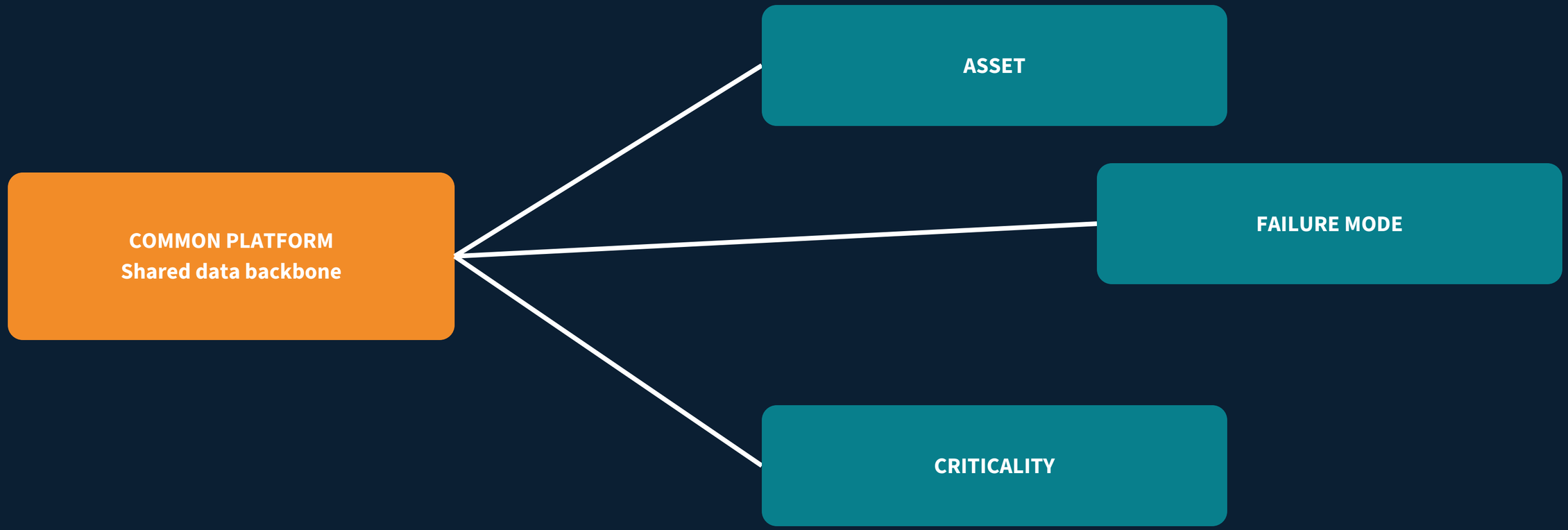
Remaining Useful Life

Insights matter only when they change a decision



A virtual replica keeps engineering context attached to sensor data—so teams can test scenarios before dispatching work.

Divergence is not fragmentation—it is fit-for-purpose design



Different assets fail in different signatures



MOTOR
Bearing wear

PUMP
Cavitation

VALVE
Leakage / stiction

Sensing strategy follows the failure hypothesis

ASSET	VIBRATION	TEMP.	CURRENT	PRESSURE	ACOUSTIC
Motor	High-rate	Slow	Medium	—	Optional
Pump	High-rate	Slow	Medium	Medium	Useful
Valve	Event	Slow	—	High	Useful

Sampling rate, placement and context are part of the model—not an afterthought.

Choose the simplest analysis that answers the decision

01 THRESHOLD

Transparent, fast; best with known limits.

02 ANOMALY

Finds departures when labels are scarce.

03 CLASSIFY

Names known failure patterns.

04 RUL

Estimates time-to-intervention.

Criticality sets the maintenance posture

HIGH CRITICALITY

Conservative thresholds • redundancy • frequent review

LOW CRITICALITY

Condition-based or reactive • low-cost sensing

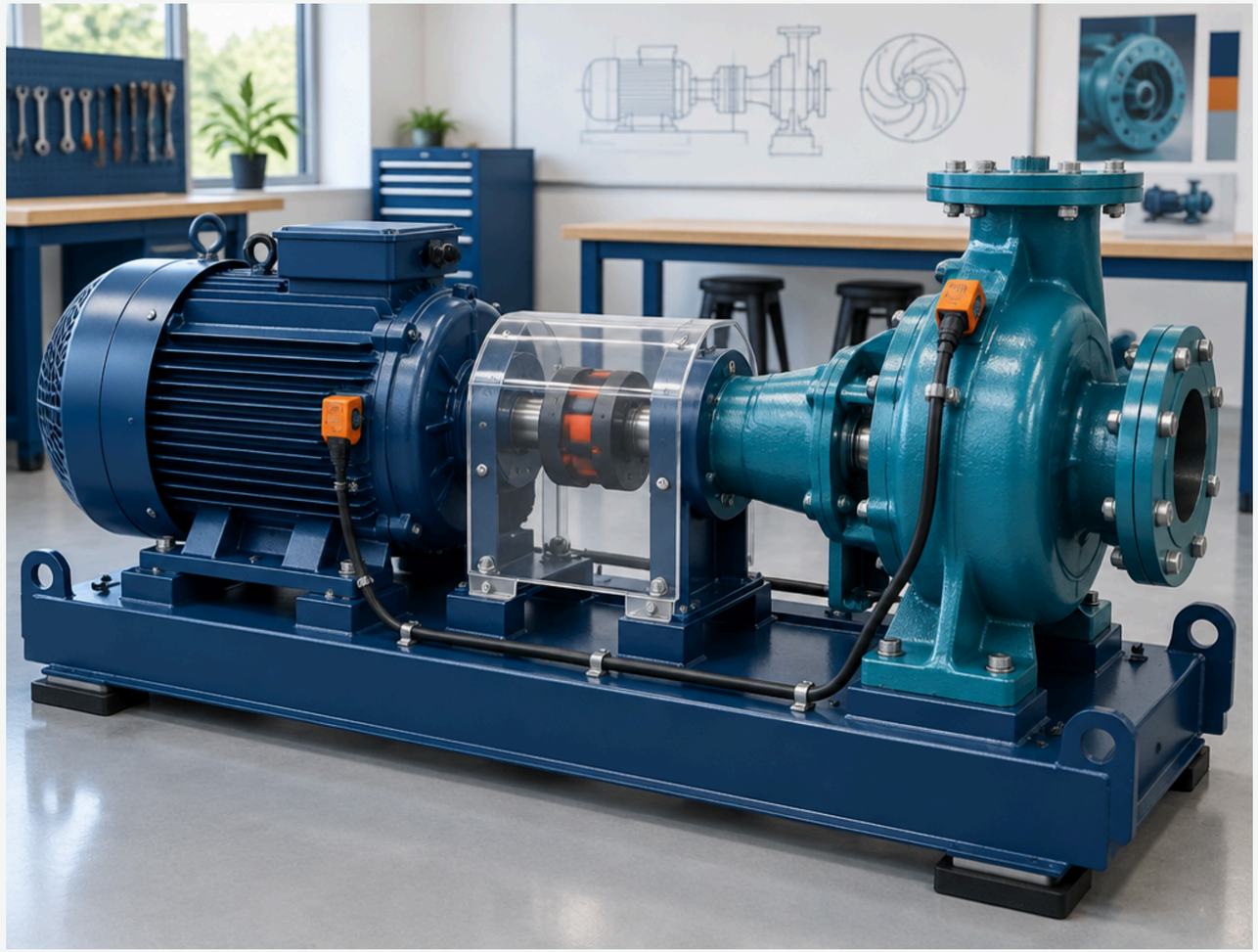
HIGH FAILURE RISK

Earlier intervention • larger safety margin

LOW FAILURE RISK

Monitor trend • defer work when evidence is weak

Use case: one drivetrain, two tailored views



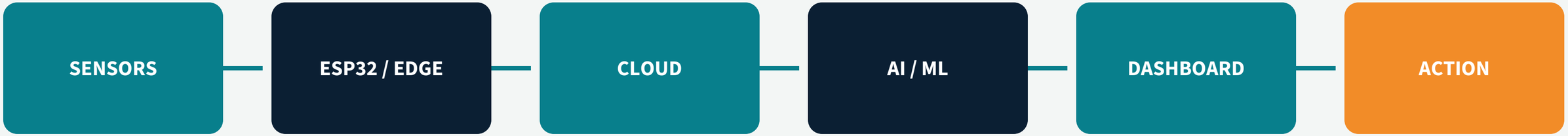
MOTOR VIEW

Vibration + current
Focus: bearing wear and imbalance

PUMP VIEW

Vibration + pressure + temperature
Focus: cavitation and flow instability

End-to-end architecture: shared spine, adaptive edges



CONVERGENCE → shared transport, identity, context and workflow

DIVERGENCE → per-asset sensors, features, models and thresholds

A learning prototype makes t

- Vibration, temperature and current sensing
- Local feature extraction
- Dashboard anomaly alerts
- Modular, low-cost experimentation

LEARNING PROTOTYPE
Not industrial safety-rated



Prototype value grows through disciplined validation

BENEFITS

Cost-effective • flexible • educational

LIMITATIONS

Sensor accuracy • robustness • prototype status

1 REFINE
Calibrate & harden

2 FIELD TEST
Compare with baseline

3 INTEGRATE
Plan industrial interfaces

Unify the platform. Tailor the maintenance decision.

CONVERGENCE creates scale.

DIVERGENCE creates fit.

Effective predictive maintenance combines a unified data-to-action backbone with asset-specific sensing, analytics and risk choices.

SELECTED REFERENCES

ISO 13374 — Condition monitoring and diagnostics

ISO 17359 — General guidelines for condition monitoring

Jardine, Lin & Banjevic (2006) — Machinery diagnostics and prognostics

Lee et al. (2014) — Cyber-physical systems for manufacturing